Metric	Level	Calculation	Interpretation
Total strength (trajectories)	Node-based metric	In-strength (sum of incoming trajectories) + out-strength (sum of outgoing trajectories). Computed for each MPA	Overall climate connectivity flow; high values indicate greater thermal turnover through an MPA, and high local connectivity with other MPAs (a hub).
Edge strength (trajectories)	Edge-based metric	Total flow between a pair of MPAs in both directions.	Overall climate connectivity between pairs of MPAs; high values indicate strong bidirectional climate connectivity.
Betweenness centrality ($\log_{10}$)	Node-based metric	Fraction of shortest paths between all node pairs passing through a focal node (Cecino & Treml 2021). Edge weights transformed to be the same rank-order as distance ((1/n) and $\log_{10}$ transformed).	Importance of each MPA in network-wide climate connectivity. High values indicate that an MPA functions as an important intermediate stepping-stone along trajectories.
MPA residence time (years)	Node-based metric	Sum of monthly time steps during which climate trajectories were present within each MPA polygon. Represented as the median across all ESMs, per SSP–period combination.	Retention of climate trajectories within an MPA, signalling the duration of climate suitability or thermal persistence—longer residence times indicate slower local climate velocity or repeated trajectory convergence.
Cumulative trajectory protection	Trajectory-based metric	Proportion of monthly time steps that trajectories have spent inside MPAs, across their 20-year lifetime. Represented as the median across all ESMs, per SSP–period combination, and aggregated across periods within each SSP.	Reflects the protection accrued as a trajectory moves through the seascape. High values indicate that a trajectory remains within MPA boundaries for most of its displacement pathway; low values indicate that trajectories spend the majority of their lifetime in unprotected areas.